

Ore mineral characteristics as rate-limiting factors in sphalerite flotation: Comparison of the mineral chemistry (iron and manganese content), grain size, and liberation

Glaciale Tiu^{a,*}, Yousef Ghorbani^b, Nils Jansson^a, Christina Wanhainen^a, Nils-Johan Bolin^c

^a Division of Geosciences and Environmental Engineering, Luleå University of Technology, Luleå SE-97187, Sweden

^b Division of Minerals and Metallurgical Engineering, Luleå University of Technology, Luleå SE-97187, Sweden

^c Boliden Mineral AB, SE-93632 Boliden, Sweden

ARTICLE INFO

Keywords:

Sphalerite
Froth flotation
Lappberget
Garpenberg mine
Automated mineralogy
QEMSCAN®

ABSTRACT

A sphalerite deportment study using QEMSCAN® was conducted to understand the distribution of the sphalerite grains with varying iron and manganese content during a multi-stage Zn-Pb-Cu flotation of a complex ore. In addition, the study aims to understand the effect of sphalerite chemistry in relation to grain size and liberation. Mineral chemistry, grain size, and degree of liberation were extracted for 497,704 sphalerite grains, composed of approximately 31 million energy-dispersive spectrometer (EDS) analysis points. Bulk chemical assay and electron probe microanalysis (EPMA) were used to validate the extracted sphalerite chemistry data from QEMSCAN®. Sphalerite recovery in the final zinc concentrate improved with increasing iron and manganese content for sphalerite grains containing up to 9% Fe and 2% Mn. At higher iron and manganese content, sphalerite recovery decreased or plateaued. Conversely, an opposite trend was observed for losses to the copper-lead concentrate. The obtained results indicated that considering the ore mineral characteristics, physical factors such as grain size and liberation have a more significant effect on sphalerite flotation than the presence of iron and manganese.

1. Introduction

Sphalerite (ZnS) is the principal ore mineral for zinc (Zn) and accounts for approximately 95% of the world's zinc mine production (International Lead and Zinc Study Group, 2021). The crystal structure of sphalerite allows it to incorporate a broad range of elements, occurring from trace (e.g., gallium (Ga), germanium (Ge), indium (In), cobalt (Co)) to minor concentrations (e.g., iron (Fe), manganese (Mn), cadmium (Cd)). The presence of these impurities can cause distortions in the sphalerite crystal structure that can affect its mineral surface properties (Babedi et al., 2021). For the majority of zinc ores, iron is the most common impurity in sphalerite, with concentrations generally ranging to levels up to 15 wt% but can be as high as 31 wt% (Di Benedetto et al., 2005; Patrick et al., 1999). Manganese-rich sphalerite is less common but has been reported from a few zinc deposits (e.g., Gottesmann and Kampe, 2013; McClung and Viljoen, 2011). Few studies have been conducted to understand the effect of impurities in sphalerite flotation, namely, on xanthate adsorption, copper activation, and consequently,

reagent consumption.

Chen et al. (2010) proposed that based on the frontier molecular orbital theory, theoretically, pure sphalerite (ZnS) will be less reactive to xanthate as compared to those with impurities (e.g. Mn, Fe, Co, Ni, Cu, Hg and Pb). However, short-chain thiol collectors such as xanthate have a relatively low affinity for zinc ions, producing an unstable and highly soluble Zn-xanthate species (Du Rietz, 1975; Patrick et al., 1999). Thus, sphalerite flotation requires copper activation of the sphalerite surface to allow xanthate adsorption. This involves the replacement of Zn^{2+} in the mineral surface by Cu^{2+} from solution, which allows the xanthate collector to react with the Cu species formed at the surface, mainly CuS (covellite) and Cu_2S (chalcocite) (Buckley et al., 1989; Ralston and Healy, 1980a). In alkaline conditions, $\text{Cu}(\text{OH})_2$ can precipitate at the sphalerite surface and may undergo ion exchange with Zn^{2+} followed by reduction to Cu^+ to form Cu(I)-xanthate and dixanthogen (Buckley et al., 1989; Popov and Vućinić, 1990; Ralston and Healy, 1980b).

Theoretical studies on the effect of iron impurities on the copper activation of sphalerite yielded conflicting results. Several studies

Abbreviations: QEMSCAN, Quantitative Evaluation Of Minerals By Scanning Electron Microscopy.

* Corresponding author.

E-mail address: glaciale.tiu@ltu.se (G. Tiu).

<https://doi.org/10.1016/j.mineng.2022.107705>

Received 21 June 2021; Received in revised form 6 June 2022; Accepted 15 June 2022

Available online 30 June 2022

0892-6875/© 2022 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

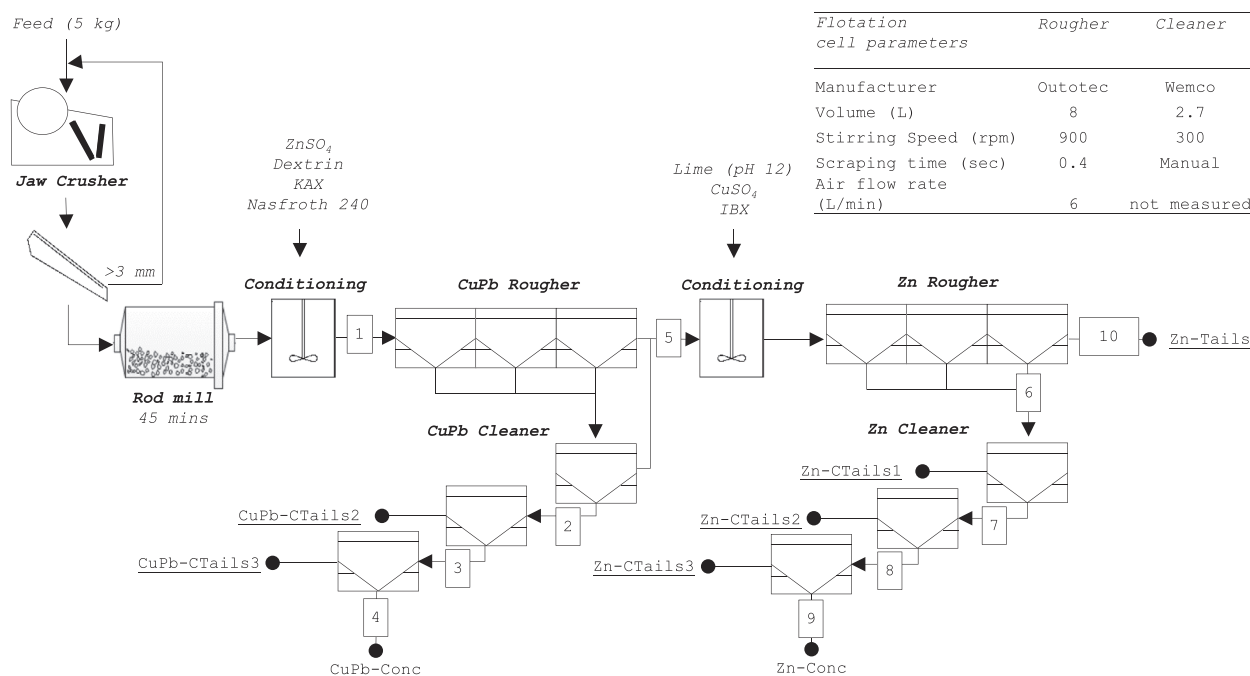


Fig. 1. Laboratory flotation test scheme. Analyzed flotation products utilized in this study are underlined whereas calculated flotation streams are numbered.

showed more efficient copper activation with increasing iron content (Gigowski et al., 1991; Harmer et al., 2008; Nefedov et al., 1980; Pat-trick et al., 1999), whereas other studies reported the opposite trend (Babedi et al., 2021; Boulton et al., 2005; Buckley et al., 1989; Chen et al., 2012; Pomianowski et al., 1975; Solecki et al., 1979; Szczypa et al., 1980). These conflicting results may be due to differences in the experimental design (e.g., pH, mineral substrate, presence of other trace element substituents, reagent regime, and other factors). Most studies also utilized monomineralic natural and synthetic sphalerite ore, making it difficult to relate the results in a real industrial context. The assumption that sphalerite in ores generally has the same composition is not valid for natural conditions, especially for complex ores. Different mineral formation conditions (i.e., temperature, pressure, and chemistry of the crystallization environment) can produce sphalerite with varying mineralogical characteristics within the same deposit (e.g., Gamsberg, South Africa (McClung and Viljoen, 2012), Hilton, Australia (Cave et al., 2020), Lappberget, Sweden (Tiu et al., 2021), Red Dog, USA (Zieliński et al., 2000)).

Compared to iron, the impact of manganese in sphalerite flotation is less understood. Flotation tests conducted by Greet et al. (2000) showed better recovery for sphalerite with higher manganese content. On the other hand, a study by Schouwstra et al. (2010) showed lower recovery for manganese-rich sphalerite ore. The results showed that the presence of alabandite (MnS) had a more significant effect on the final grade and recovery due to copper being preferentially adsorbed on the alabandite mineral surface rather than the sphalerite. There is still a lack of understanding on the effect of manganese content in sphalerite during copper activation.

Economically, the impact of impurities in sphalerite is reflected on the reagent consumption. With the current copper price (ca. 10,000 USD/ton; London Metal Exchange average price for May 2022), copper (II) sulfate ($CuSO_4$) is one of the most expensive flotation reagent in the Zn-Pb-Cu flotation process. The increased copper ion coverage on the sphalerite surface due to higher iron content will, in effect, require a higher dosage of copper activator to achieve optimal grade and recovery (Harmer et al., 2008; Nefedov et al., 1980; Pat-trick et al., 1999). For example, it was reported in the Laisvall mine in Sweden (operated by Boliden Mineral AB) that ores with iron-rich sphalerite (3% Fe)

consumed more $CuSO_4$ than ores containing low iron content (1.5% Fe). In comparison between the two operating zinc mines by Boliden Mineral AB, $CuSO_4$ consumption does not vary significantly between the Tara Mine in Ireland and the Garpenberg mine despite the former having mostly iron-poor sphalerite ($Fe\% < 0.5$). However, it is difficult to constrain the effect of impurities in sphalerite given that the two mentioned Boliden mines process ores from different sources.

To partially fill the knowledge gap, a deportment study was conducted on a complex ore with sphalerite of varying iron and manganese content. The present contribution aims to understand the effect of the presence of iron and manganese impurities in sphalerite during a multi-stage Zn-Pb-Cu flotation under similar conditions as a typical mineral processing plant. Automated mineralogy (QEMSCAN®, Gottlieb et al., 2000) was utilized to extract the individual composition and properties of sphalerite grains from the flotation samples and validated using electron probe microanalysis (EPMA). The significance of iron and manganese impurities in sphalerite was presented in terms of sphalerite recovery for the different flotation stages and related to grain size and degree of liberation. From the extracted sphalerite particle data, multiple linear regression modelling was conducted for sphalerite recovery prediction as a function of the iron and manganese content, grain size, and degree of liberation.

2. Methodology

2.1. Ore sample

For this study, approximately 22 meters of quartered drill core samples were collected from the skarn-hosted sulfide mineralization from the Lappberget Zn-Pb-Ag-(Cu-Au) deposit, which is currently exploited by Boliden Mineral AB's Garpenberg mine in Sweden (see SKN mineralization from Tiu et al., 2021). The samples mainly consisted of sphalerite, pyrite, and galena, which were hosted in skarn-altered manganoan dolomitic marble. The samples were selected due to the abundance of sphalerite and the presence of sphalerite grains with varying iron and manganese contents, as previously identified by Tiu et al. (2021). To produce a 5-kg feed for the flotation test, the samples were crushed to less than 3.15 mm using a jaw crusher then

Table 1

Factors utilized to calculate sphalerite recovery for data analysis and modelling.

Factors	Categories	Value	Interval
Fe content	11	0 to > 10%	1%
Mn content	11	0 to > 5%	0.50%
Grain size	11	0 to > 50 μm	5 μm
Degree of Liberation	5	0 to 100%	20%

Table 2

Summary of the sphalerite composition based on the EPMA data.

	S (%)	Zn (%)	Fe (%)	Mn (%)	Cd (%)
Min	32.50	53.20	2.40	0.10	0.10
Max	33.60	63.80	9.40	5.20	0.40
Average	32.92	60.29	5.67	0.90	0.21
Std. Dev.	0.21	2.06	1.44	0.74	0.05

homogenized and split using a rotary sample splitter.

2.2. Flotation test

A multi-stage batch flotation test was conducted, including a 45-min grinding stage using a rod mill followed by a copper-lead flotation stage and then a zinc flotation stage. Fig. 1 shows the comminution and flotation procedures followed in this study.

Two flotation tests were conducted to duplicate the experiment. Zinc sulfate (ZnSO_4) and dextrin (Dextrin Gul) were used in the copper-lead rougher flotation stage to depress sphalerite whereas potassium amyl xanthate (KAX) and triethylene glycol monobutyl ether (Nasfroth 240) were used as a collector and frother reagent, respectively. Copper (II) sulfate (CuSO_4) was used as an activator before the Zn rougher flotation stage whereas isobutyl xanthate (IBX) was used as collector. Slaked lime ($\text{Ca}(\text{OH})_2$) was added to adjust the pH value to 12. Eight flotation products were collected as marked in Fig. 1. The characteristics of the flotation streams (e.g., feed) were derived based on the analyzed flotation products as listed in Appendix A.

2.3. Mineralogical characterization

The flotation products from the two flotation tests were dried, weighed, split and sent to Intertek Genalysis Laboratory in Australia for chemical analyses. The samples were dissolved using four acid digestion and analyzed for 48 elements using inductively coupled plasma (ICP)–mass spectrometer (MS) and optical emission spectrometer (OES) (see Intertek (2020) for a more detailed description of the procedures 4A/MS48 and 4AH/OE). Since there is no significant difference between the chemical assay results for the two flotation tests (see Appendix B), mineralogical characterization was only performed for the flotation products of flotation Test 1.

Modal mineralogy, composition, and particle characteristics were analyzed using QEMSCAN® 650 (FEI with W-filament, two EDS (Energy-Dispersive Spectrometer), and an electron backscatter detector) at Boliden Mineral's laboratory. Prior to the analysis, the flotation samples were dried and sized into three fractions using sieves 45 and 63 μm . Polished resin mounts were prepared for the different size fractions (i.e., <45, 45–63, >63 μm) and sputtered with carbon to provide a conductive coating. The particle mineralogical analysis (PMA) measurement mode was performed with an emission current of 10 nA and an accelerating voltage of 25 kV. The extent of each particle is determined based on pre-defined electron backscatter threshold (set above the backscatter coefficient of the epoxy resin). Spectra were collected for the entire area of the particle at different spatial resolution: 2.2 μm (<45 μm size fraction), 3.3 μm (45–63 μm), and 4.8 μm (>63 μm). Each spectrum contains a minimum of 1000 X-ray counts. The species identification protocol (SIP) or mineral library used to discriminate minerals were modified based on

the mineralogical characterization and composition by Tiu et al. (2021). FEI™ iMeasure version 5.4 and iDiscover 5.4 software were used for data collection and processing.

Electron probe microanalysis (EPMA) was performed on 153 randomly selected sphalerite grains from the three flotation products (Zn-Conc, Zn-CTails2 and Zn-Tails). The samples were analyzed using an electron microprobe (Jeol SuperProbe JXA-8230) equipped with five wavelength-dispersion spectrometers (PETJ, PETH, PETL, LIF, LIFH, LIFL, TAP, LDE1 and TAPH) at AGH University of Science of Technology, Poland. The sphalerite analyses were determined using an accelerating voltage of 20 kV with a probe current and beam diameter of 20 nA and 1 μm , respectively. Natural minerals and metals were used as reference standards. Counting times of 20 s on peak and 10 s on both (+) and (–) backgrounds were used. The following standards and spectral lines were used: sphalerite ($\text{SK}\alpha$, $\text{ZnK}\alpha$), chalcopyrite ($\text{CuK}\alpha$), CdS ($\text{CdL}\alpha$), pyrite ($\text{FeK}\alpha$) and MnS ($\text{MnK}\alpha$). Data was corrected to the ZAF procedure using JEOL software. Data for the EPMA spot analyses is attached as supplementary material (Appendix E).

2.4. Data extraction

QEMSCAN® pixel data was extracted and analyzed in Microsoft® SQL Server and ioGAS. The composition of sphalerite grains was calculated based on the average composition for each mineral grain from the individual EDS results from the QEMSCAN® analysis. EDS data with less than 100,000 X-ray count rate (kcps) per spectrum and 85% spectral engine identification confidence was removed to reject outliers and mixed analyses. Due to the detection limit of the EDS analysis, only the concentrations of zinc, iron, manganese and sulfur were taken into account. Sphalerite composition was normalized based on the EDS data for zinc, iron and manganese, and a value of 33 wt% for sulfur and 0.2 wt% for cadmium (based on the average data from EPMA). For the particle size measurement, virtual sieving was conducted based on the equivalent circular area diameter (ECD). Mass was calculated based on the area percentage of the analyzed mineral grains and the density of minerals as listed in the QEMSCAN® Mineral Database v.50.1.1.1.

2.5. Data analysis

Principal component analysis (PCA) and multiple linear regression model were fitted for the calculated sphalerite recovery as a function of the iron and manganese content, grain size, and degree of liberation of the sphalerite-bearing particles. The zinc content of sphalerite was not included since the amount of zinc is collinear with the amount of iron and manganese in sphalerite (i.e., higher iron and manganese content will result to lower zinc content). The linear model utilized in this study followed this notation:

$$\text{Sphalerite Recovery}(\%) \sim \text{Fe}(\%) + \text{Mn}(\%) + \text{Grainsize}(\mu\text{m}) + \text{Degree of liberation}(\%) + \varepsilon \quad [1]$$

wherein *Sphalerite Recovery* was calculated based on the mass of sphalerite grains in the final concentrate with respect to the feed. The “ ε ” is an error term that accounts for all other parameters that can affect sphalerite recovery that is either random or uncontrollable. Sphalerite recovery data was generated by creating subsets of sphalerite-bearing particle data from the different combinations of the four factors listed in Table 1. The data was randomly split into a training (80%) and testing (20%) data set. The linear model was fitted against the training data and then validated using the testing data. The absence of collinearity was assumed for the four factors to create the linear model. A more detailed discussion on the statistical requirements for creating linear models can be found in van den Boogaart and Tolosana-Delgado (2013) and Winter (2013). Linear modelling and data analysis were done using R software suite version 4.1.0 (R Core Team, 2021) with lm function (Chambers, 1992; Wilkinson and Rogers, 1973). An example of the script is given in

Table 3

Modal mineralogy of the feed based on the QEMSCAN® analysis.

Mineral	Weight %
Sphalerite	47.8
Pyrite	22.6
Silicates	15.7
Galena	10.7
Carbonates	4.2
Pyrrhotite	0.4
Oxides	<0.1
Chalcopyrite	<0.1
Others	<0.1

Appendix C.

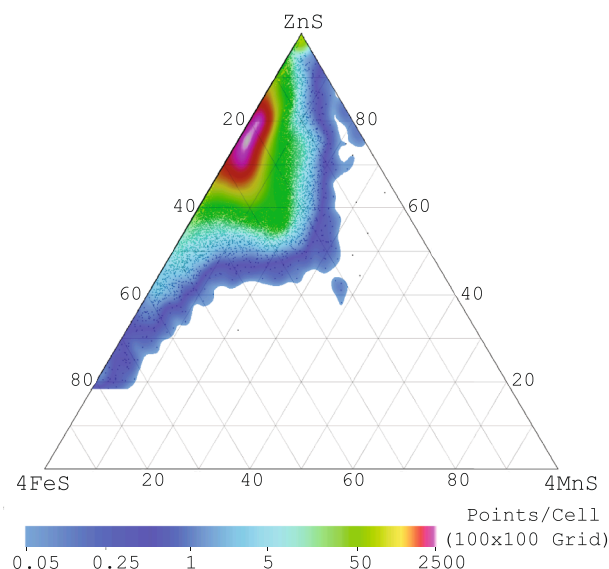
3. Results and discussion**3.1. Sphalerite occurrence**

Based on the EPMA data, significant variation in iron and manganese content in sphalerite is observed as summarized in Table 2 and provided as a [supplementary material \(Appendix E\)](#). Sulfur and cadmium content is relatively constant at 33 wt% and 0.2 wt%, respectively. The result agrees with the more extensive mineralogical study by Tiu et al. (2021) wherein three sphalerite types with increasing Mn/Fe ratios and distinct trace elements were distinguished. Sphalerite-1 is generally iron-rich (ca. 0.1 Mn/Fe) and coarse grained whereas sphalerite-2 has higher Mn content (ca. 0.4 Mn/Fe). The last type of sphalerite (sphalerite-3) is fine-grained and occurs in veins characterized by high manganese content (>1 Mn/Fe). The collected samples in this study contain all three types of sphalerite based on the QEMSCAN® analysis, wherein majority is sphalerite-1 type. Only sphalerite-1 and sphalerite-2 were identified in the selected grains analyzed in the EPMA.

Textural differences are also observed between the sphalerite types, which reflect the crystallizing conditions during the formation of sphalerite crystals. The ore utilized in this study, which is from the Lappberget deposit, have undergone peak metamorphic condition of up to 550 °C at pressures below 3.5 kbar (Vivallo, 1985). During high-temperature metamorphic conditions, grain growth can occur during annealing and recrystallization. In addition, sulfide inclusions within sphalerite (commonly chalcopyrite and pyrrhotite) migrate along grain boundaries. Both mechanisms produce coarse-grained and liberated sphalerite as observed for most sphalerite-1 grains, which has relatively higher iron content compared to sphalerite-2 (Tiu et al., 2021). Conversely, natural “milling” of sulfides can occur within mineralized shear zones during low-temperature brittle deformation. This produces “ball ore texture” characterized by a fine-grained sulfide matrix, which in the case in Lappberget, mainly consists of sphalerite and galena, with rounded gangue clasts.

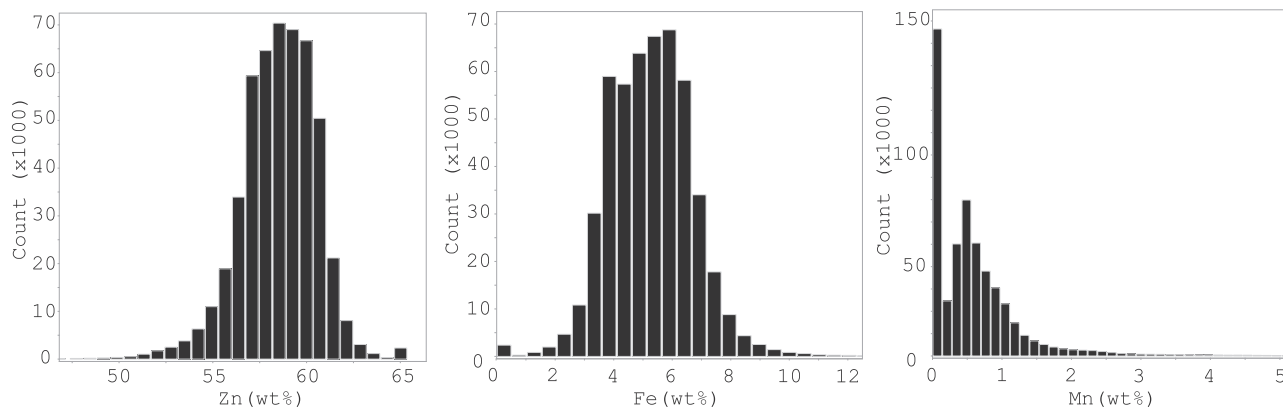
3.2. Data statistics

A total of 497,704 sphalerite grains, composed of approximately 31 million EDS analysis points, were classified based on the variations in their mineral chemistry. Modal mineralogy of the feed was calculated based on the QEMSCAN® analysis and is provided in Table 3. The composition of sphalerite varies significantly in terms of iron and manganese content as shown in the histograms and ternary plot in Figs. 2 and 3. The sphalerite types are not clearly distinguished in the extracted data, possibly due to significant abundance of sphalerite-1,

**Fig. 3.** Ternary plot illustrating the sphalerite compositional variation in ZnS-FeS-MnS space. Color scale represents point density.**Table 4**

Measured and calculated chemical assays in wt% of the flotation products.

Flotation Products	Chemical Assay			QEMSCAN®		
	Mn	Fe	Zn	Mn	Fe	Zn
CuPb-Conc	0.2	3.6	11.0	0.1	3.7	4.4
CuPb-CTails2	0.3	18.2	9.4	0.2	20.2	5.2
CuPb-CTails3	0.3	13.0	11.1	0.2	15.0	7.1
Zn-Conc	1.0	5.8	60.1	0.9	5.9	58.7
Zn-CTails1	1.0	6.9	61.6	1.1	7.1	56.5
Zn-CTails2	1.3	5.9	45.6	0.9	6.2	51.8
Zn-CTails3	1.1	5.7	55.9	0.8	5.6	58.4
Zn-Tails	1.8	11.7	1.5	2.3	15.1	1.9

**Fig. 2.** Histograms of Zn, Fe, and Mn content in sphalerite.

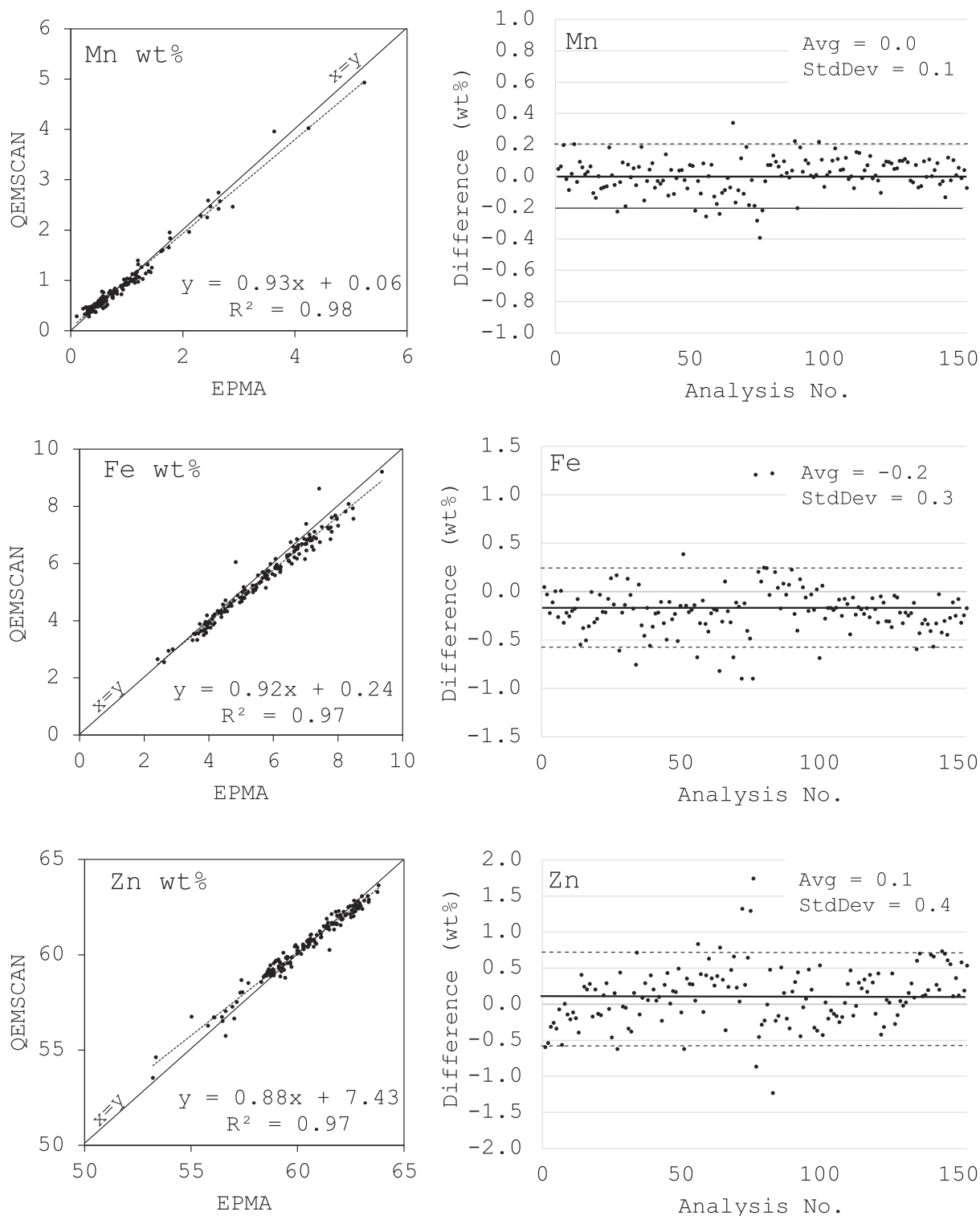


Fig. 4. Correspondence between EPMA and QEMSCAN® data for Mn, Fe and Zn. (Left) Comparison of EPMA and QEMSCAN® data for Mn, Fe and Zn. (Right) Differences between QEMSCAN® and EPMA data showing the average difference in thick solid line and 95% confidence interval line in dash line.

masking the presence of other populations. Median sphalerite composition is calculated at 58.6% Zn, 5.3% Fe, and 0.5% Mn resulting in an empirical formula of $(\text{Zn}_{0.9}\text{Fe}_{0.1}\text{Mn}_{0.01})\text{S}$. Iron and manganese content in sphalerite is not correlated ($r = 0.2$). Non-collinearity of iron and manganese agrees with the study by [Patrick et al. \(1998\)](#), which showed that manganese substitution in sphalerite is independent of iron content. Alternatively, the data distribution is more likely affected by the presence of different sphalerite types with varying iron and

manganese ratios ([Tiu et al., 2021](#)).

3.3. Data quality

The extracted QEMSCAN® data was validated based on two aspects: (1) the comparability of the measured (ICP-MS) and calculated (QEMSCAN®) bulk chemistry of the flotation products and (2) the correspondence between measured (EPMA) and calculated

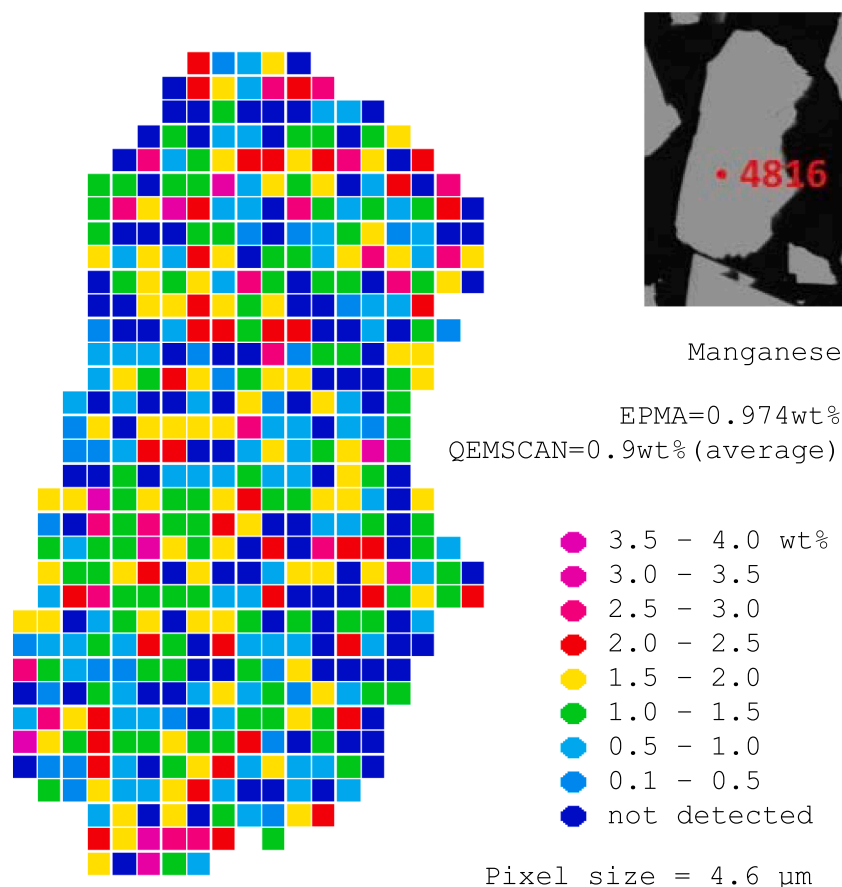


Fig. 5. Example of a particle analyzed with QEMSCAN® using EDS mapping. SEM image of the particle shows the EPMA analysis point location.

(QEMSCAN®) sphalerite chemistry.

3.3.1. Bulk chemistry

The analyzed manganese, iron and zinc content of the flotation products was compared against the calculated grade from the QEMSCAN® analysis (Table 4). Significant differences between the two analytical methods are observed, which is expected given the differences on the sampling procedure and uncertainty of QEMSCAN® as compared with the chemical analysis by ICP-MS. In addition, mineral chemistry data was only corrected and optimized for sphalerite (e.g. Fe- and Mn-bearing amphibole mineral chemistry was based on average mineral composition). Nonetheless, the results for manganese, iron and zinc are relatively comparable especially for the sphalerite-rich flotation products (Zn-flotation stage).

3.3.2. Sphalerite chemistry

A comparison of the QEMSCAN® analysis and EPMA data for manganese, iron and zinc is shown in Fig. 4 and provided as a [supplementary material](#) (Appendix E). Despite the semi-quantitative nature of QEMSCAN® in terms of mineral chemistry, good correspondence between the two data sets is observed, i.e., strong positive linear relationship and high R^2 . For manganese, iron and zinc, 95% confidence interval of the QEMSCAN® data falls within ± 0.2 , ± 0.6 , and ± 0.8 wt% of the EPMA values, respectively. QEMSCAN® measurements for iron are slightly underestimated especially for higher values whereas the opposite is observed for zinc, i.e., slightly overestimated for lower values (Fig. 4). For iron, only two particles have greater than 1 wt% difference between the two datasets. These are sphalerite grains locked with pyrite grains (<20% liberation). Although the average sphalerite chemistry closely approximates the EPMA analysis, individual EDS points from the QEMSCAN® analysis show high variability (e.g. Fig. 5).

3.4. Flotation performance

3.4.1. Mineral chemistry

Sphalerite recovery was calculated for all flotation stages as a function of sphalerite composition (Fig. 6). Sphalerite recovery for the rougher and the first and second cleaning stages exhibit similar trends. As shown in Fig. 6A, significantly lower sphalerite recovery is observed for sphalerite with low iron and manganese content for the final zinc concentrate. This indicates that iron- and manganese-poor sphalerite-bearing particles are lost to the cleaner tails during the final cleaning stage.

In terms of the final zinc concentrate, sphalerite recovery increases with increasing Fe + Mn content in sphalerite (Fig. 6A). Individually, an increased sphalerite recovery is observed with increased iron content in sphalerite (up to 6% Fe) then plateaus and decreases at higher iron content (>9% Fe) (Fig. 6B). A similar trend is observed for manganese, that is, increased sphalerite recovery up to 2% which then decreases at higher grades (>5% Mn). Highest sphalerite recovery is observed for sphalerite grains with approximately 1:2 Mn/Fe ratio, which decreases significantly for grains with high Mn/Fe ratio (>0.9 Mn/Fe) (Fig. 6D). Sphalerite recovery for the copper-lead concentrate shows a reversed pattern of that of the zinc concentrate, i.e., iron- and manganese-poor sphalerite preferentially misreporting to the copper-lead concentrate.

3.4.2. Grain size

The grain size distribution of analyzed sphalerite grains yields a P_{80} value of 37 μm (ECD). Sphalerite with low iron content (<1%) is distinctly finer-grained than the other types (Fig. 7A). Iron-poor sphalerite, which was mostly classified as sphalerite-2 in Tiu et al. (2021), is generally finer-grained than sphalerite-1 as it has not undergone intensive recrystallization associated with grain growth. The presence of

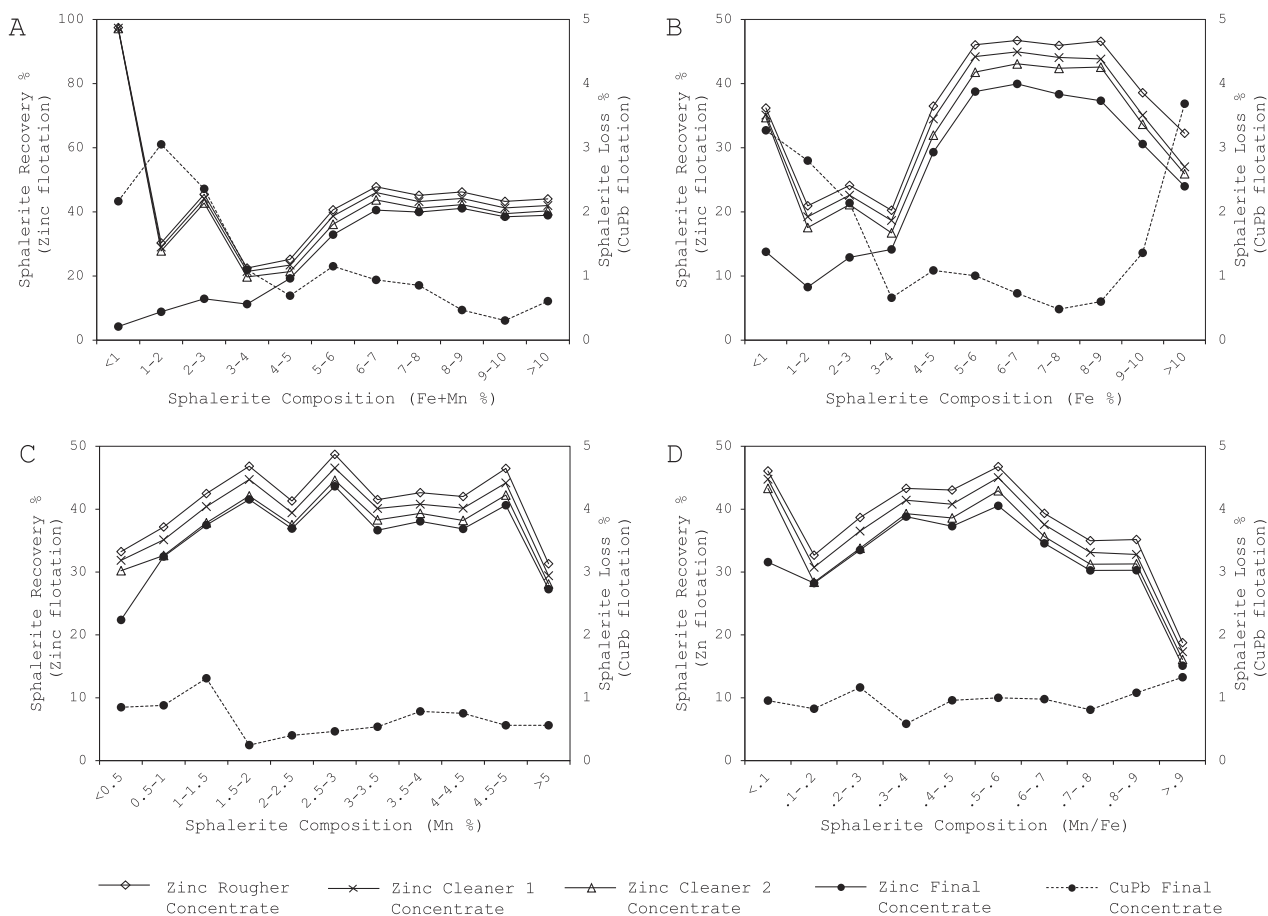


Fig. 6. Sphalerite recovery/loss for the different flotation stages versus sphalerite composition in terms of (A) Fe + Mn, (B) Fe, (C) Mn and (D) Mn/Fe content.

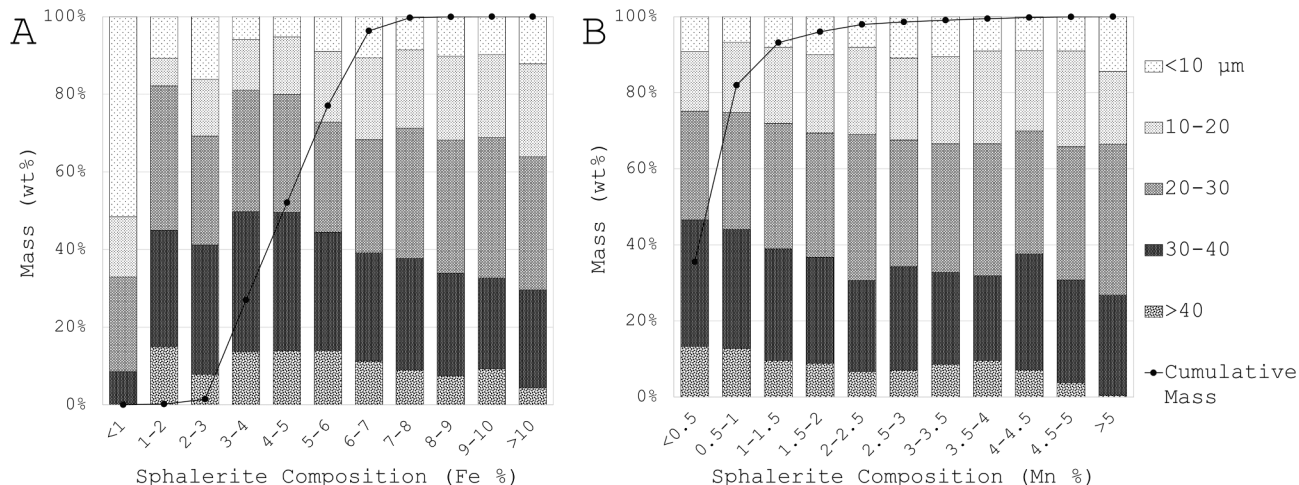


Fig. 7. Mass distribution of sphalerite grains for different size classes in terms of (A) Fe and (B) Mn content of sphalerite in the feed.

iron-poor fine-grained sphalerite (<10 μm) possibly explains the sphalerite particles that were subsequently dropped out during the final cleaning stage (Fig. 6). Fine-grained particles have low particle momentum, which makes them easier to be washed out with water (Pease et al., 2006). Fig. 7 shows a slight decrease in the amount of coarse sphalerite grains (>40 μm) with increasing iron and manganese content.

Sphalerite recovery as a function of iron and manganese content in sphalerite and grain size are shown in Fig. 8. It is evident that sphalerite recovery in the final zinc concentrate decreases with increasing grain

size. This is possibly due to the higher probability of detachment for large particles during froth flotation (Reay and Ratcliff, 1975; Trahar, 1981). Sphalerite recovery in terms of iron and manganese content shows similar trends for the different grain size classes. This indicates that the effect of the presence of impurities in sphalerite on sphalerite recovery is a mineralogical factor independent of grain size.

3.4.3. Liberation

Up to 86% of the analyzed sphalerite grains are well liberated (>80%

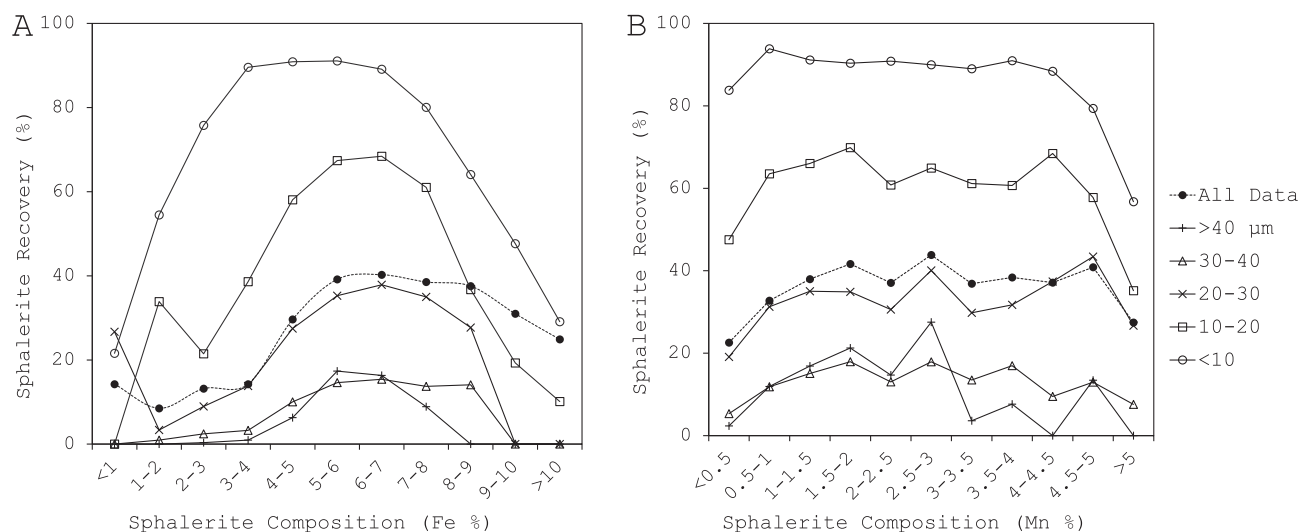


Fig. 8. Sphalerite recovery based on the final zinc concentrate for the different sphalerite grain size classes in terms of (A) Fe and (B) Mn content.

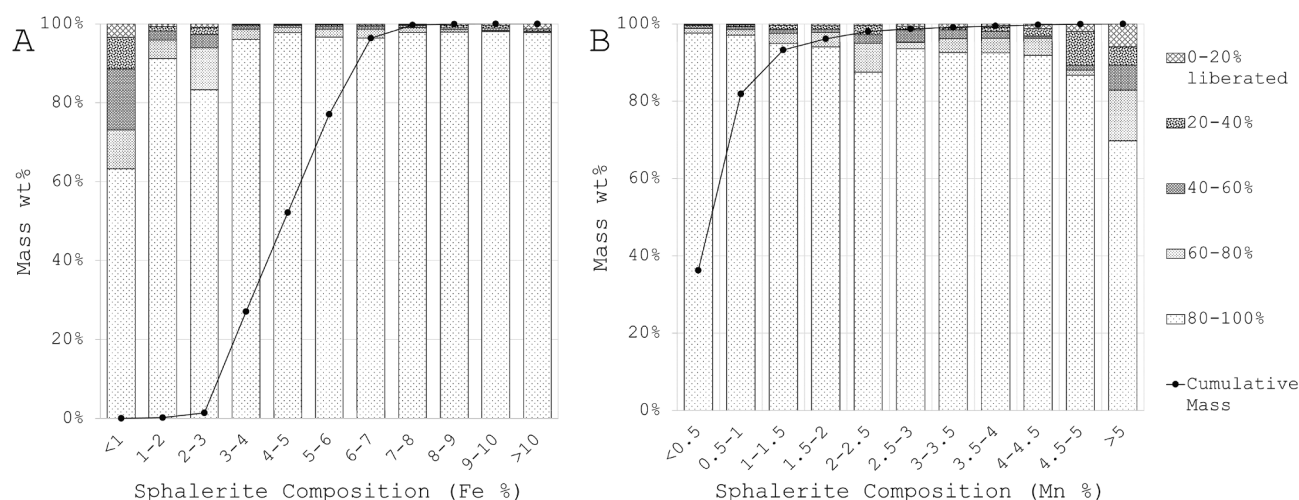


Fig. 9. Mass distribution of sphalerite grains in the feed as a function of the degree of liberation and (A) Fe and (B) Mn content of sphalerite.

liberation). Sphalerite grains with low iron content show a lower degree of liberation whereas an inverse trend is observed in terms of manganese content (Fig. 9). The presence of non-liberated particles could have affected the sphalerite recovery, which may be another reason for the low sphalerite recovery in the final zinc concentrate for sphalerite grains with low iron and high manganese content (Fig. 6B and 6C). Poorly liberated particles are expected to have a slow flotation rate or none at all (Feng and Aldrich, 1999; Zhang, 1989).

To remove the effect of non-liberated particles, sphalerite recovery is plotted for well-liberated particles (>80% liberation) in Fig. 10. An increasing trend in sphalerite recovery with iron and manganese content is still observed. Since the majority of the analyzed sphalerite grains are well liberated, the plotted sphalerite recovery for all data generally follows the same trend for both zinc and copper-lead concentrates. For iron-rich sphalerite (>8%), the effect of non-liberated particles causes a slight decrease in the overall trend for sphalerite recovery (Fig. 10B).

Based on the liberated particles' performance, increased floatability for sphalerite grains is still observed with increasing iron content in the zinc flotation circuit. This result is conflicting with two of the three published studies conducted over the past three decades on the effect of sphalerite iron content on copper activation using natural sphalerite samples under alkaline pH regime (pH 10–12) (Table 5). Boulton et al. (2005) and Zhang et al. (2021) reported that iron is detrimental to the

copper activation of sphalerite and, in effect, yielded lower sphalerite recovery. However, Patrick et al. (1999) suggested that even though the presence of iron in sphalerite inhibits the initial copper activation, it catalyzes covellite formation upon xanthate addition with a rate faster than pure sphalerite. In addition, Harmer et al. (2008) suggested that iron enhances copper adsorption due to the increased number of surface defects in the sphalerite structure and increased size of surface oxidation products. Iron is preferentially removed from the surface due to faster surface diffusion and a more stable configuration than zinc. Results from Zhang et al. (2021) showed that although copper activation decreases with increasing iron content, xanthate adsorption only started to decrease for sphalerite samples containing 8.7% Fe and higher. Boulton et al. (2005) also utilized a highly iron-rich sphalerite sample (12.5% Fe). Although in general, this study showed an increased sphalerite recovery with increasing iron content, a slight decrease in sphalerite recovery was observed with sphalerite containing more than 9% Fe (Fig. 10B). Thus, the results from this study is not necessarily contradictory from previous studies, but rather points to the possibility of a 'critical' iron content in sphalerite wherein sphalerite recovery is optimal. However, the absence of a significant amount of well-liberated, extremely iron-rich sphalerite grains makes it difficult to conclusively confirm and extend this decreasing trend. The lack of studies conducted with a continuous and wide range of sphalerite iron content may also

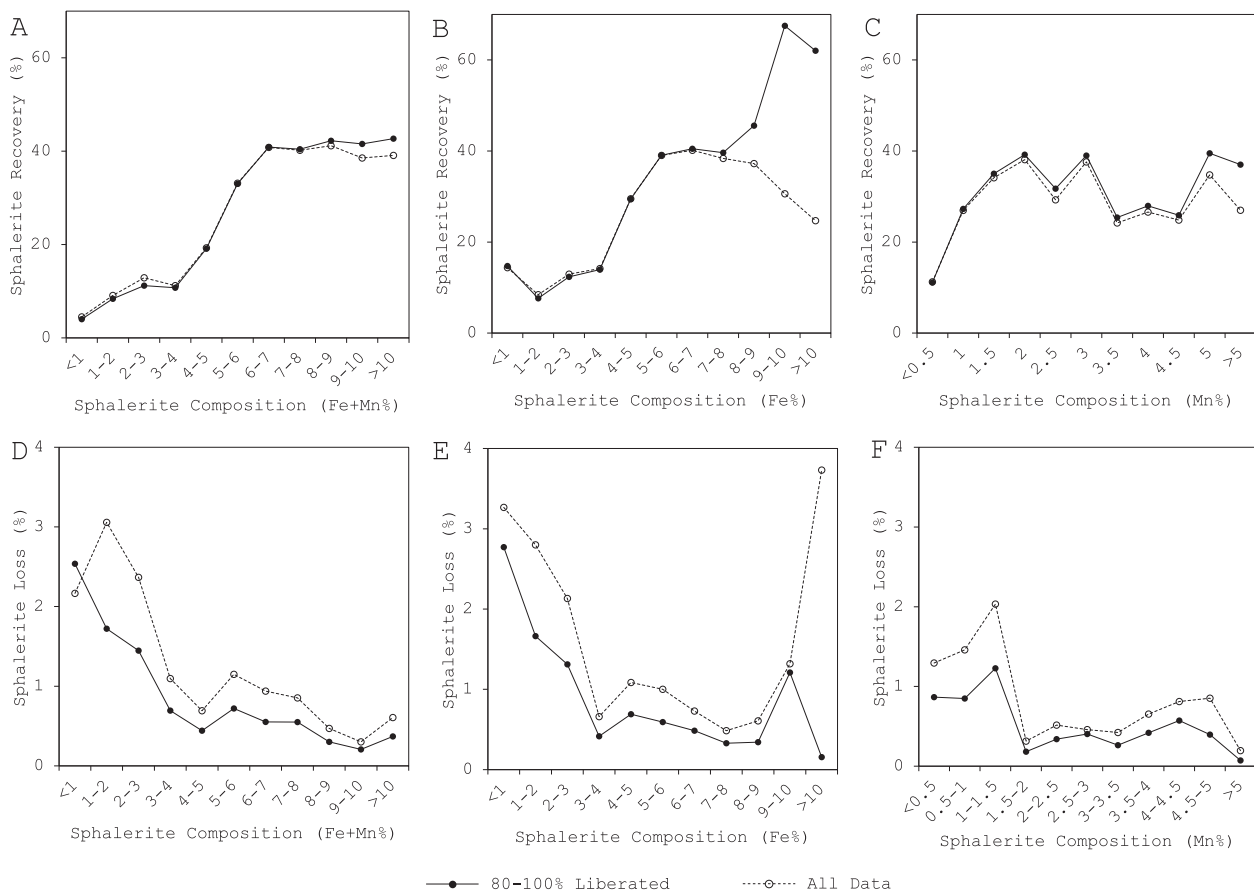


Fig. 10. Sphalerite recovery based on the final zinc concentrate (A-C) and sphalerite losses to the final copper-lead concentrate (D-F) for well-liberated particles (80–100% liberation) and all data with respect to (A, D) Fe + Mn, (B, E) Fe, and (C, F) Mn content in sphalerite.

Table 5

Flotation and copper activation studies using natural ores at alkaline pH regimes.

Fe content in sphalerite (%)	Size (μm)	Activator	Collector	pH	Reference
0.3 (natural)	P ₉₀ = 45	CuSO ₄	sodium isopropyl	11	Boulton et al. (2005)
12.5			xanthate		
1.1 (natural)	38–74	CuSO ₄ ·5H ₂ O	potassium n-butyl xanthate	12	Zhang et al. (2021)
4.9					
8.7					
16.2					
0.025 (synthetic)	<10 (synthetic) not given (natural)	CuSO ₄	sodium isopropyl	10 and 12	Patrick et al. (1999)
14.6					
0 (synthetic)					
2.3					
4.5					

* Copper activation studies were only conducted for natural ore samples

explain the contradictory results from previous studies.

Similar to iron, sphalerite recovery increases up to 2% Mn, which becomes less pronounced at higher grades. The increased recovery with manganese content in sphalerite is also observed by Greet et al., (2000). However, there is still a lack of copper activation studies on the behavior of manganese in sphalerite during sphalerite flotation.

Relatively high sphalerite losses to the copper-lead concentrate is attributed to iron- and manganese-poor sphalerite in well-liberated particles (Fig. 10D to F). It is worth noting that well-liberated sphalerite grains misreporting to the copper-lead concentrate only account for less than 0.6 wt% of sphalerite in the feed (2.5 wt% if only considering the near-stoichiometrically pure sphalerite (Fe + Mn) < 1%). Several

authors have also reported minute amounts of iron-poor sphalerite or near-stoichiometrically pure sphalerite (ZnS) misreporting to the copper and lead concentrate (Finkelstein, 1997; Mellberg, 1979; Morey et al., 2001; Ralston and Healy, 1980a, 1980b; Trahar et al., 1997; Zieliński et al., 2000). Previous studies suggested that this is due to the inadvertent activation of sphalerite by Cu²⁺ (Finkelstein, 1997) and Pb²⁺ ions (Rashchi et al., 2002; Trahar et al., 1997) from impurities in the sphalerite lattice, process water, or the associated ore minerals. Both the EPMA results from this study and from Tiu et al. (2021) show no significant copper and lead content within the sphalerite. Since the water utilized in the flotation test is not recycled, it is most likely that the ions come from the associated ore minerals, i.e., dissolution and/or oxidation

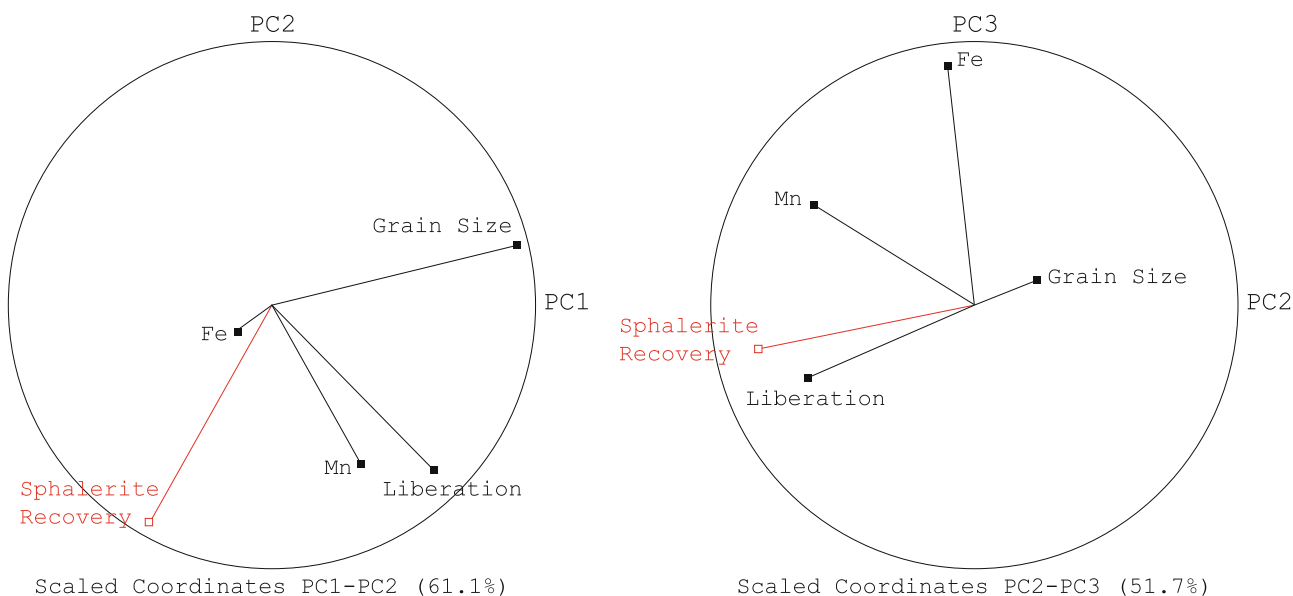


Fig. 11. PCA loading plots (PC1 vs PC2 and PC2 vs PC3 vector plane) based on the sphalerite recovery in the final zinc concentrate.

Table 6

Statistical parameters of the factors extracted from the multiple linear regression model for sphalerite recovery prediction.

Factors/Coefficients	Estimate	Std. Error	p-value
Fe content	2.05	0.48	2.04e-5
Mn content	1.67	0.81	0.04
Grain Size	-1.71	0.06	< 2e-16
Degree of Liberation	0.77	0.04	< 2e-16
(Intercept)	-2.39	4.48	0.59

of chalcopyrite and galena during the grinding and flotation process. Copper activation by adsorption of Cu^{2+} ions is similar to the effect of CuSO_4 in the zinc flotation process. Lead activation is through Pb substitution for Zn in the sphalerite lattice under acidic conditions, formation of Zn-O-Pb^+ species on the surface through $\text{Pb}(\text{OH})^+$ adsorption at pH 7–10, and lead hydroxide precipitation on the sphalerite surface at pH > 10 (Rashchi et al., 2002). The addition of ZnSO_4 helps inhibit the adsorption of copper and lead ions on the sphalerite surface as well as inhibit collector attachment in the copper-lead flotation process. However, the efficiency of lead activation and the depressing action of ZnSO_4 in the presence (or absence) of impurities in sphalerite remains poorly understood.

3.5. Statistical analysis

Principal component analysis of the sphalerite recovery with respect to iron and manganese content, grain size, and degree of liberation confirms the findings shown in the previous sections. The loading plots of the principal components 1 to 3 (PC1-PC3) contributing to 83% data variance show a strong negative influence of grain size with sphalerite recovery (Fig. 11). Iron and manganese content and degree of liberation show a positive correlation (refer to Appendix D for a more detailed PCA analysis report).

The result of the multiple linear regression modelling using the training data set is summarized in Table 6. Validation result between the predicted and experimental sphalerite recovery of the test data is plotted in Fig. 12. The extremely low p-value of the linear model ($p < 2e-16$) confirms its significance; however, the residual standard error is found to be high at 20.77, suggesting that other factors (e.g., random effects) should be accounted to improve the model. Nonetheless, the linear model confirms the significance of the four individual factors (p-value < 0.05) (Table 6). The extremely low p-value for sphalerite grain size suggests a strong influence on sphalerite recovery followed by the degree of liberation and iron content. A slightly higher p-value for manganese can be due to the insufficient sample population for sphalerite containing high manganese content. The correlation between sphalerite recovery and the other factors is reflected in the estimated coefficients for each factor in Table 6 (i.e., negative for grain size and positive for Fe content, Mn content, and degree of liberation). Boulton et al. (2005) also showed that the rate of flotation with different iron composition was significantly affected by grain size. Sphalerite grain size in the feed is within the optimal range for froth flotation, i.e. between 10 and 50 μm

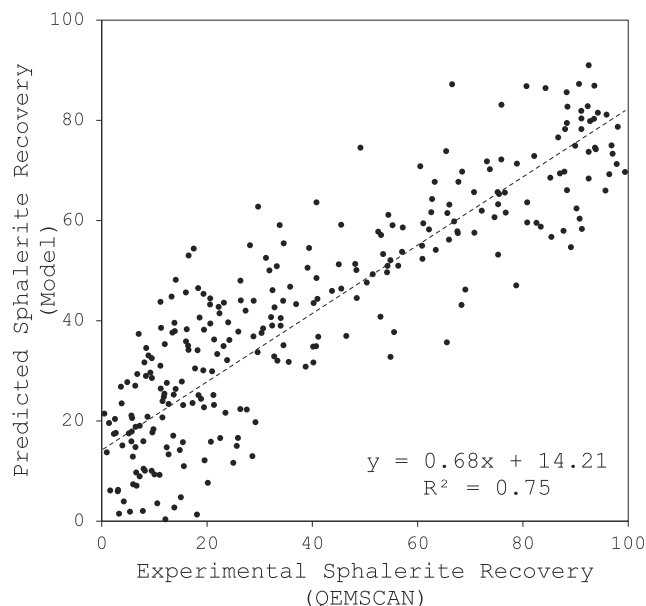


Fig. 12. Comparison between the experimental and predicted values of the test data set based on the multiple linear regression model for sphalerite recovery in the final zinc concentrate.

(Anthony et al., 1975; Feng and Aldrich, 1999; Gaudin et al., 1931; Reay and Ratcliff, 1975; Trahar, 1981; Zhang, 1989).

Absence of collinearity between the four factors was assumed in the linear model, which as shown in this study is not true in nature especially for complex ores. In addition, the model is not significantly affected by sphalerite with extremely high iron and manganese content and coarse grain size (i.e. $> 40 \mu\text{m}$) due to limited data. In effect, the decrease in sphalerite recovery observed for extremely iron- and manganese-rich sphalerite is not accounted in the model. Thus, this linear model cannot be used to model flotation kinetics, but can be utilized to understand the significance of the mineralogical factors for flotation.

4. Conclusion

Quantitative mineralogical characterization using automated mineralogy (QEMSCAN®) provided a comprehensive sphalerite dataset, which was utilized to determine the effect of impurities such as iron and manganese in sphalerite during the multi-stage Zn-Pb-Cu flotation process. The effect of grain size and degree of liberation with respect to sphalerite mineral chemistry was also presented. The following major conclusions were derived from the presented results:

- Sphalerite recovery in the final zinc concentrate increases with increasing iron content (up to 9% Fe) then decreases at higher grades. A similar trend is also observed for sphalerite containing up to 2% Mn, which becomes less pronounced at higher grades. Conversely, majority of the well-liberated sphalerite grains lost to the copper-lead concentrate have low iron and manganese content.
- Grain size and degree of liberation have a more significant effect than iron and manganese content in sphalerite in terms of sphalerite recovery. Grain size shows a negative correlation with sphalerite recovery whereas the opposite is observed for the degree of liberation.
- Multiple linear regression modelling of the sphalerite recovery as a function of iron and manganese content, grain size, and degree of liberation confirmed that all factors have a significant effect on sphalerite recovery. Additional flotation results from different ore samples with wider sphalerite impurities and grain size distribution are necessary to constrain both fixed and random effects needed in producing a robust mathematical model to be applied for a flotation kinetics study.

Appendix A. Derivation formulas for the flotation streams

No.	Flotation Stream	Composition Calculation
1	Feed	Sum of all flotation products
2	CuPb Cleaner 1 Conc	$\text{CuPb-CTails2} + \text{CuPb-CTails3} + \text{CuPb-Conc}$
3	CuPb Cleaner 2 Conc	$\text{CuPb-CTails3} + \text{CuPb-Conc}$
4	CuPb Final Conc	CuPb-Conc
5	CuPb Tails	$\text{Zn-CTails1} + \text{Zn-CTails2} + \text{Zn-CTails3} + \text{Zn-Conc} + \text{Zn-Tails}$
6	Zn Rougher Conc	$\text{Zn-CTails1} + \text{Zn-CTails2} + \text{Zn-CTails3} + \text{Zn-Conc}$
7	Zn Cleaner 1 Conc	$\text{Zn-CTails2} + \text{Zn-CTails3} + \text{Zn-Conc}$
8	Zn Cleaner 2 Conc	$\text{Zn-CTails3} + \text{Zn-Conc}$
9	Zn Final Conc	Zn-Conc
10	Final Tails	Zn-Tails

Automated mineralogy such as QEMSCAN® was shown to be effective for successful discrimination of sphalerite based on iron and manganese content. This method offers an attractive solution in understanding the deportment of minor elements within the flotation process, but as well can be utilized to produce spatially resolved data for geological and geometallurgical purposes. However, for the positive detection of minor elements, it is important to acquire the data at high resolution and optimal equipment setting. Quantitative chemical and mineral analyses (e.g. EPMA) are equally important to validate the data.

CRediT authorship contribution statement

Glaciale Tiu: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft, Validation, Visualization. **Yousef Ghorbani:** Methodology, Writing – review & editing, Supervision. **Nils Jansson:** Methodology, Writing – review & editing, Supervision. **Christina Wanhainen:** Conceptualization, Methodology, Writing – review & editing, Supervision, Funding acquisition, Resources, Project administration. **Nils-Johan Bolin:** Methodology, Writing – review & editing.

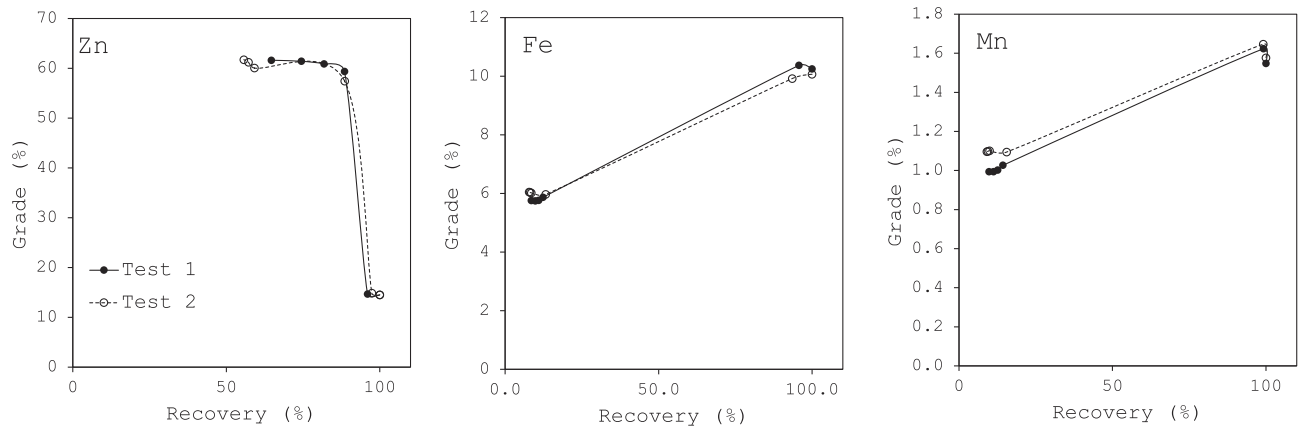
Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This study was financed by the European Union's Horizon 2020 research and innovation program under grant agreement No. 72267, as part of the MetalIntelligence network (www.metalintelligence.eu). The authors would like to acknowledge Boliden Mineral AB for providing us technical and financial support. The authors are highly grateful to the Boliden geologists, engineers and technicians, especially Iris Mc Elroy, who have provided their knowledge and time to assist us during this study. The authors would also like to thank Dr. Gabriela Kozub-Budzyńis for performing the EPMA analysis and FEI (iDiscover) for granting access to the software utilized in this study. This paper greatly benefited from the reviews by the two anonymous reviewers.

Appendix B. Grade and recovery curves for Zn, Fe and Mn



Appendix C. R code

Import data as data.frame containing values for sphalerite recovery, Fe, Mn, grain size and liberation. The linear model “mdl” will be computed based on Equation 1. The error term is not necessary for the lm() function.

```
> Mdl <- lm(SphaleriteRecovery ~ Fe + Mn + GrainSize + Liberation, data = data).
```

To get a summary of the model, use function summary():

```
> summary(Mdl).
```

To check for the linearity and normality of the residuals, the following functions can be used:

```
> plot(fitted(Mdl),residuals(Mdl)).
```

```
> hist(residuals(Mdl)).
```

```
> qqnorm (residuals(Mdl)).
```

Appendix D. PCA analysis report

Transformation None, Scaling True.

Correlation	GrainSize	Liberation	Fe	Mn	Sphalerite Recovery
GrainSize	1	0.3874	-0.02	0.113	-0.5189
Liberation	0.3874	1	-0.128	0.2879	0.2943
Fe	-0.02	-0.128	1	0.1415	0.05158
Mn	0.113	0.2879	0.1415	1	0.1638
Sphalerite Recovery	-0.519	0.2943	0.0516	0.1638	1

Eigenvectors	PC1	PC2	PC3	PC4	PC5
GrainSize	0.7388	0.1849	0.0822	0.1858	0.6154
Liberation	0.4891	-0.515	-0.26	0.3992	-0.5183
Fe	-0.105	-0.085	0.8744	0.4545	-0.1028
Mn	0.2709	-0.498	0.367	-0.738	-0.001944
SphaleriteRecovery	-0.361	-0.668	-0.162	0.2357	0.5849
PC1	Eigenvalues	Percent			Cumulative %
PC1	1.555	31.09			31.09
PC2	1.5	29.99			61.08
PC3	1.086	21.72			82.81
PC4	0.6761	13.52			96.33
PC5	0.1835	3.671			100

Scaled Coordinates	PC1	PC2	PC3	PC4	PC5
GrainSize	0.9212	0.2264	0.0856	0.1528	0.2636
Liberation	0.6099	-0.631	-0.271	0.3283	-0.2221
Fe	-0.131	-0.104	0.9113	0.3738	-0.04402
Mn	0.3378	-0.61	0.3825	-0.607	-8.33E-04
SphaleriteRecovery	-0.451	-0.817	-0.169	0.1938	0.2506

Appendix E. EPMA and QEMSCAN results in wt% for selected sphalerite grains

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.mineng.2022.107705>.

References

- Anthony, R.M., Kelsall, D.F., Trahar, W.J., 1975. The effect of particle size on the activation and flotation of sphalerite. In: *Proceedings of the Australian Institute of Mining and Metallurgy*, pp. 47–58.
- Babedi, L., Tadie, M., Neethling, P., von der Heyden, B.P., 2021. A fundamental assessment of the impacts of cation (Cd Co, Fe) substitution on the molecular chemistry and surface reactivity of sphalerite. *Miner. Eng.* 160, 106695. <https://doi.org/10.1016/j.mineng.2020.106695>.
- Boulton, A., Fornasier, D., Ralston, J., 2005. Effect of iron content in sphalerite on flotation. *Miner. Eng.* 18 (11), 1120–1122. <https://doi.org/10.1016/j.mineng.2005.03.008>.
- Buckley, A.N., Woods, R., Wouterlood, H.J., 1989. An XPS investigation of the surface of natural sphalerites under flotation-related conditions. *Int. J. Miner. Process.* 26 (1–2), 29–49. [https://doi.org/10.1016/0301-7516\(89\)90041-0](https://doi.org/10.1016/0301-7516(89)90041-0).
- Cave, B., Lilly, R., Hong, W., 2020. The effect of co-crystallising sulphides and precipitation mechanisms on sphalerite geochemistry: a case study from the Hilton Zn-Pb (Ag) deposit, Australia. *Minerals* 10, 797. <https://doi.org/10.3390/min10090797>.
- Chambers, J.M., 1992. Linear Models, in: *Statistical Models in S*. p. 50.
- Chen, Y.e., Chen, J., Guo, J., 2010. A DFT study on the effect of lattice impurities on the electronic structures and floatability of sphalerite. *Miner. Eng.* 23 (14), 1120–1130. <https://doi.org/10.1016/j.mineng.2010.07.005>.
- Chen, Y., Chen, J., Lan, L., Yang, M., 2012. The influence of the impurities on the flotation behaviors of synthetic ZnS. *Miner. Eng.* 27–28, 65–71. <https://doi.org/10.1016/j.mineng.2012.01.001>.
- Di Benedetto, F., Bernardini, G.P., Costagliola, P., Plant, D., Vaughan, D.J., 2005. Compositional zoning in sphalerite crystals. *Am. Mineral.* 90, 1384–1392. <https://doi.org/10.2138/am.2005.1754>.
- Du Rietz, C., 1975. Chemisorption of collectors in flotation. *Proc. XI Int. Min. Proc. Congress.* 375–403.
- Feng, D., Aldrich, C., 1999. Effect of particle size on flotation performance of complex sulphide ores. *Miner. Eng.* 12 (7), 721–731. [https://doi.org/10.1016/S0892-6875\(99\)00059-X](https://doi.org/10.1016/S0892-6875(99)00059-X).
- Finkelstein, N.P., 1997. The activation of sulphide minerals for flotation: a review. *Int. J. Miner. Process.* 52 (2–3), 81–120. [https://doi.org/10.1016/S0301-7516\(97\)00067-7](https://doi.org/10.1016/S0301-7516(97)00067-7).
- Gaudin, A.M., Groh, J.O., Henderson, H.B., 1931. Effect of particle size on flotation. *Am. Inst. Min. Metall. Eng. Tech. Publ.* 414, 3–23.
- Gigowski, B., Vogg, A., Wierer, K., Dobias, B., 1991. Effect of Fe-lattice ions on adsorption, electrokinetic, calorimetric and flotation properties of sphalerite. *Int. J. Miner. Process.* 33 (1–4), 103–120. [https://doi.org/10.1016/0301-7516\(91\)90046-L](https://doi.org/10.1016/0301-7516(91)90046-L).
- Gottesmann, W., Kampe, A., 2013. The Tumurtijn-ovoo Fe-Mn-Zn skarn deposit, Mongolia: sedimentary features, distribution of main minerals and the whole-rock chemical composition. *Neues Jahrb. fur Mineral. Abhandlungen* 191 (1), 1–32. <https://doi.org/10.1127/0077-7757/2013/0243>.
- Gottlieb, P., Wilkie, G., Sutherland, D., Ho-Tun, E., Suthers, S., Perera, K., Jenkins, B., Spencer, S., Butcher, A., Rayner, J., 2000. Using quantitative electron microscopy for process mineralogy applications. *JOM* 52 (4), 24–25. <https://doi.org/10.1007/s11837-000-0126-9>.
- Greet, C., Butcher, A.R., Gottlieb, P., 2000. Quantification of High- and Low-Manganese Sphalerite from Dugald River Flotation Concentrates Using QEMSCAN, in: *Flotation 2000*.
- Harmer, S.L., Mierczynska-Vasilev, A., Beattie, D.A., Shapter, J.G., 2008. The effect of bulk iron concentration and heterogeneities on the copper activation of sphalerite. *Miner. Eng.* 21 (12–14), 1005–1012. <https://doi.org/10.1016/j.mineng.2008.02.014>.
- International Lead and Zinc Study Group, 2021. *World Directory: Lead and Zinc Mines*. Intertek, 2020. Mineral Services: Schedule of Services and Charges Australia.
- McClung, C.R., Viljoen, F., 2012. Mineralogical Assessment of the Metamorphosed Broken Hill Sulfide Deposit, South Africa: Implications for Processing Complex Orebodies, in: Broekmans, M.A.T.M. (Ed.), *Proceedings of the 10th International Congress for Applied Mineralogy (ICAM)*. Springer Berlin Heidelberg, pp. 427–434. https://doi.org/10.1007/978-3-642-27682-8_51.
- McClung, C.R., Viljoen, F., 2011. A detailed mineralogical assessment of sphalerites from the Gamsberg zinc deposit, South Africa: The manganese conundrum. *Miner. Eng.* 24 (8), 930–938. <https://doi.org/10.1016/j.mineng.2011.04.010>.
- Mellberg, F., 1979. Operating experiences from autogeneous grinding at Zinkgruvan. In: *Autogeneous Grinding Seminar*. Trondheim, Norway, pp. 1–12.
- Morey, M.S., Grano, S.R., Ralston, J., Prestidge, C.A., Verity, B., 2001. The electrochemistry of PbII activated sphalerite in relation to flotation. *Miner. Eng.* 14 (9), 1009–1017. [https://doi.org/10.1016/S0892-6875\(01\)00108-X](https://doi.org/10.1016/S0892-6875(01)00108-X).
- Nefedov, V.I., Salyn, Y.V., Solozhenkin, P.M., Pulatov, G.Y., 1980. X-ray photoelectron study of surface compounds formed during flotation of minerals. *Surf. Interface Anal.* 2 (5), 170–172. <https://doi.org/10.1002/sia.740020503>.
- Patrick, R.A.D., England, K.E.R., Charnock, J.M., Mosselmans, J.F.W., 1999. Copper activation of sphalerite and its reaction with xanthate in relation to flotation: An X-ray absorption spectroscopy (reflection extended X-ray absorption fine structure) investigation. *Int. J. Miner. Process.* 55 (4), 247–265. [https://doi.org/10.1016/S0301-7516\(98\)00036-2](https://doi.org/10.1016/S0301-7516(98)00036-2).
- Patrick, R.A.D., Mosselmans, J.F.W., Charnock, J.M., 1998. An x-ray adsorption of doped sphalerite. *Eur. J. Mineral.* 10, 239–249.
- Pease, J.D., Curry, D.C., Young, M.F., 2006. Designing flotation circuits for high fines recovery. *Miner. Eng.* 19 (6–8), 831–840. <https://doi.org/10.1016/j.mineng.2005.09.056>.
- Pomianowski, A., Szczypa, J., Poling, G.W., Leja, J., 1975. Influence of Iron Content in Sphalerite–Marmatite on Copper Ion Activation in Flotation, in: *11th International Mineral Processing Congress*. Cagliari, pp. 639–653.
- Popov, S.R., Vućinić, D.R., 1990. The ethylxanthate adsorption on copper-activated sphalerite under flotation-related conditions in alkaline media. *Int. J. Miner. Process.* 30 (3–4), 229–244. [https://doi.org/10.1016/0301-7516\(90\)90017-S](https://doi.org/10.1016/0301-7516(90)90017-S).
- R Core Team, 2021. R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria [WWW Document]. URL <https://www.r-project.org/> (accessed 2.1.21).
- Ralston, J., Healy, T.W., 1980a. Activation of zinc sulphide with CuII, CdII and PbII: I. Activation in weakly acidic media. *Int. J. Miner. Process.* 7 (3), 175–201. [https://doi.org/10.1016/0301-7516\(80\)90016-2](https://doi.org/10.1016/0301-7516(80)90016-2).
- Ralston, J., Healy, T.W., 1980b. Activation of zinc sulphide with CuII, CdII and PbII: II. Activation in neutral and weakly alkaline media. *Int. J. Miner. Process.* 7 (3), 203–217. [https://doi.org/10.1016/0301-7516\(80\)90017-4](https://doi.org/10.1016/0301-7516(80)90017-4).
- Rashchi, F., Sui, C., Finch, J.A., 2002. Sphalerite activation and surface Pb ion concentration. *Int. J. Miner. Process.* 67 (1–4), 43–58. [https://doi.org/10.1016/S0301-7516\(02\)00005-4](https://doi.org/10.1016/S0301-7516(02)00005-4).
- Reay, D., Ratcliff, G.A., 1975. Collision Model of Fine Particle Flotation 53.
- Schouwstra, R., de Vaux, D., Hey, P., Malysiak, V., Shackleton, N., Bramdeo, S., 2010. Understanding Gamsberg - A geometallurgical study of a large stratiform zinc deposit. *Miner. Eng.* 23 (11–13), 960–967. <https://doi.org/10.1016/j.mineng.2010.03.013>.
- Solecki, J., Komosa, A., Szczypa, J., 1979. Copper ion activation of synthetic sphalerites with various iron contents. *Int. J. Miner. Process.* 6 (3), 221–228. [https://doi.org/10.1016/0301-7516\(79\)90038-3](https://doi.org/10.1016/0301-7516(79)90038-3).
- Szczypa, J., Solecki, J., Komosa, A., 1980. Effect of surface oxidation and iron contents on xanthate ions adsorption of synthetic sphalerites. *Int. J. Miner. Process.* 7 (2), 151–157. [https://doi.org/10.1016/0301-7516\(80\)90007-1](https://doi.org/10.1016/0301-7516(80)90007-1).
- Tiu, G., Jansson, N., Wanhainen, C., Ghorbani, Y., Lilja, L., 2021. Ore mineralogy and trace element (re)distribution at the metamorphosed Lappberget Zn-Pb-Ag-(Cu-Au) deposit, Garpenberg, Sweden. *Ore Geol. Rev.* 135, 104223. <https://doi.org/10.1016/j.oregeorev.2021.104223>.
- Trahar, W.J., 1981. A rational interpretation of the role of particle size in flotation. *Int. J. Miner. Process.* 8 (4), 289–327. [https://doi.org/10.1016/0301-7516\(81\)90019-3](https://doi.org/10.1016/0301-7516(81)90019-3).
- Trahar, W.J., Senior, G.D., Heyes, G.W., Creed, M.D., 1997. The activation of sphalerite by lead – a flotation perspective. *Int. J. Miner. Process.* 49 (3–4), 121–148.
- van den Boogaart, K.G., Tolosana-Delgado, R., 2013. *Analyzing Compositional Data with R*. Springer Berlin Heidelberg, Berlin, Heidelberg. <https://doi.org/10.1007/978-3-642-36809-7>.
- Vivallo, W., 1985. The geology and genesis of the Proterozoic massive sulfide deposit at Garpenberg, central Sweden. *Econ. Geol.* 80, 17–32. <https://doi.org/10.2113/gsecongeo.80.1.17>.
- Wilkinson, G.N., Rogers, C.E., 1973. Symbolic description of factorial models for analysis of variance. *Appl. Stat.* 22 (3), 392. <https://doi.org/10.2307/2346786>.
- Winter, B., 2013. Linear models and linear mixed effects models in R with linguistic applications. *arXiv* 1–42.
- Zhang, J.-G., 1989. *Factors affecting the kinetics of froth flotation*. University of Leeds.
- Zhang, S., Deng, Z., Xie, X., Tong, X., 2021. Study on the depression mechanism of calcium on the flotation of high-iron sphalerite under a high-alkalinity environment. *Miner. Eng.* 160, 106700. <https://doi.org/10.1016/j.mineng.2020.106700>.
- Zieliński, P.A., Larson, K.A., Stradling, A.W., 2000. Preferential deportment of low-iron sphalerite to lead concentrates. *Miner. Eng.* 13 (4), 357–363. [https://doi.org/10.1016/S0892-6875\(00\)00018-2](https://doi.org/10.1016/S0892-6875(00)00018-2).